Jeremy Hui

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Research Interests

I am interested in the intersection between **computer architecture** and **security**, especially in memory systems security such as Trusted Execution Environments (TEEs), physical attacks (e.g. side-channel attacks, Rowhammer), confidential computing, and privacy of non-volatile memory.

Education

Rutgers University – Engineering Honors Academy (GPA: 4.00/4.00) — Sep. 2022 – May 2026
B.S. Double Major in Electrical/Computer Engineering and Computer Science, Minor in Math — *New Brunswick, NJ*
- **Honors:** Tau Beta Pi, Sophia Sheng Endowed Scholarship, Dean's List, The Dean's Scholarship
- **Relevant Courses:** Hardware & Systems Security, Computer Architecture & Assembly Language, Operating Systems, Network-Centric Programming, Complexity Theory, Linear Systems & Signals, Electronic Devices

Experience

HWSEL - Hardware Security Research — Sept. 2024 – Jul. 2025
Research Assistant — *New Brunswick, NJ*
- Explored disaggregated memory through leveraging hardware-level isolation on Arm CCA processors.
- Designed an agent to automate deployment of Confidential Containers (CoCo) for DNNs and detect malware.
- Deployed CoCo without confidential hardware, and analyzed performance counters for indicators of compromise.

Princeton-Intel Research Experience for Undergraduates — Jun. 2024 – Jul. 2024
Summer Research Intern — *Princeton, NJ*
- Investigated secure memory through testing Rowhammer-like attacks on OpenSSD 3D NAND Flash Memory.
- Identified a reverse mapping between aggressor and victim pages using C/microcode, and demonstrated the ability of directing 50,000 bit flips to specific cells per 1 million repeated write cycles.
- Experimented with and documented how BCH error correction codes on NAND Flash Memory can be bypassed.

Wireless Information Network Laboratory (WINLAB) — Jun. 2023 – Aug. 2023
Summer Research Intern — *North Brunswick, NJ*
- Continued work on an experimentation platform about human interactivity with indoor spaces using Python and Linux, and effectively communicated results in 5-minute weekly presentations.
- Set up 25 Raspberry Pis to securely communicate environment data to be processed in Flask database by an LLM.
- Implemented Precision Time Protocol (PTP) to synchronize clocks on a nanosecond scale.

Rutgers University Course Staff — Sep. 2023 – Present
Head TA for Intro to CS, Learning Assistant for Intro to Computers for Engineers — *New Brunswick, NJ*
- Wrote 5 Java assignments and 2 auto-graders for 1000+ students promoting humanitarian applications of CS.
- Supported over 600 students through holding office hours, and coordinated logistics for 30+ TAs.
- Taught 2 group of 40+ students MATLAB through weekly 80-minute active-learning recitations.

Projects

Concurrent Web Proxy Server | *C* — Feb. 2025 – Apr. 2025
- Developed a web proxy that forwards browser requests with a multithread and a multi-process implementation.
- Created logging features for the proxy, with synchronization primitives and thread safety to prevent corruption.
- Implemented HTTPS tunneling for CONNECT requests to relay data without interpreting encrypted data.

PC Composer | *Node.js, SQLite3, React.js, AWS S3, OpenAI, Express.js* — Sept. 2024 – Dec. 2024
- Built a full stack web application to generate PC builds using an AI chatbot from 100+ computer parts.
- Established design patterns and object-oriented system specifications according to business goals/use cases in order to effectively coordinate an 8-member team. Applied an Agile development methodology.

MUSIK | *Python, PyGame, OpenCV, Unity* — Jan. 2024 – May 2025
- Engineered prototype to teach piano and enhance experience through descending notes in an AR MIDI visualizer.
- Utilized computer vision for visualizer alignment, and Bluetooth protocols to transmit playing information.

Technical Skills

Languages: HTML, JavaScript, Java, C/C++, MATLAB, Python, SQL (MySQL, SQLite3), RISC-V
Frameworks: React, Node.js, WordPress, Express.js, Jasmine
Tools: AWS EC2, gem5, Linux, GitHub, Docker, AWS S3, Vim, Vivado/Verilog, VSCode